

systems is crucial for a holistic view of your data. Explore tools like:

- **AWS Glue:** A serverless ETL service offered by Amazon Web Services that simplifies data extraction, transformation, and loading tasks on AWS.
- **Azure Data Factory:** A cloud-based ETL/ELT service from Microsoft Azure that allows data integration across various cloud and on-premises sources.
- **Google Cloud Dataflow:** A managed service for creating data pipelines on Google Cloud Platform. It supports a variety of data sources and processing options.
- **Informatica Cloud Data Integration (ICDI):** A cloud-native ETL/ELT platform offering pre-built connectors, data quality tools, and workflow automation capabilities.

Data security

Protecting your data is paramount. Implement robust security measures like:

- **Encryption:** Encrypt data at rest and in transit using cloud provider services like AWS KMS or Azure Key Vault.
- **Identity and access management (IAM):** Utilise IAM controls to restrict data access to authorised

Benefits of cloud data management

Benefits	Description
• Scalability and flexibility	Easily scale storage and processing capabilities according to business needs.
• Cost efficiency	Eliminate the need for expensive on-premises hardware; pay for resources used.
• Enhanced security	Advanced security features like encryption, access controls, and regular audits.
• Improved collaboration	Access data from anywhere, at any time, and on any device, fostering collaboration among teams.
• Disaster recovery and continuity	Robust backup and recovery mechanisms for quick data recovery and minimal downtime.

- personnel only.
- **Regular security audits:** Conduct periodic security assessments to identify and address vulnerabilities.

Data backup and recovery

Data backup involves creating copies of data to protect against loss, stored separately for redundancy. Here are key considerations for data backup.

- **Backup frequency:** Determine how often you need to back up your data based on its criticality and how frequently it changes. Real-time backups might be necessary for highly sensitive data, while less critical data can be backed up daily or weekly.
- **Backup location:** Don't store your backups in the same location

- as your primary data. Consider geographically dispersed cloud storage or on-premises backup solutions for added protection.
- **Data encryption:** Encrypt your backups at rest and in transit to safeguard sensitive information from unauthorised access, even in the event of a security breach.
 - **Backup testing:** Regularly test your backups to ensure they are complete and readily recoverable. This provides peace of mind knowing you can restore your data promptly in case of an incident.
- Data recovery is the process of restoring data from backups in case of loss or corruption. It ensures business continuity, data integrity, and compliance with regulations. Here's what a robust data recovery plan entails:
- **Backup cataloguing:** Maintain a clear catalogue of your backups, including information like creation date, data source, and location. This ensures you can quickly identify the right backup for recovery.
 - **Point-in-time recovery:** Some backup solutions allow you to recover data to a specific point in time. This is valuable when you need to restore data from a specific moment before an error or attack occurred.
 - **Disaster recovery plan:** Develop a comprehensive disaster recovery plan that outlines procedures for responding to various data loss scenarios. This plan should

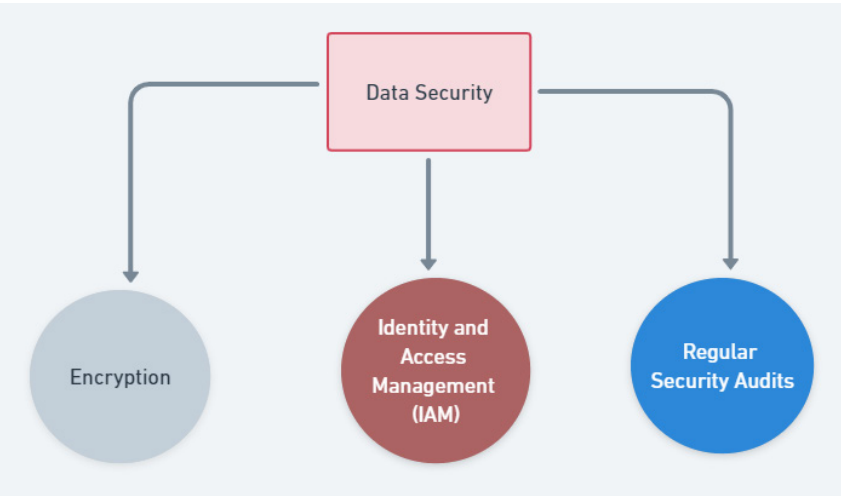


Figure 4: Data security